

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 18

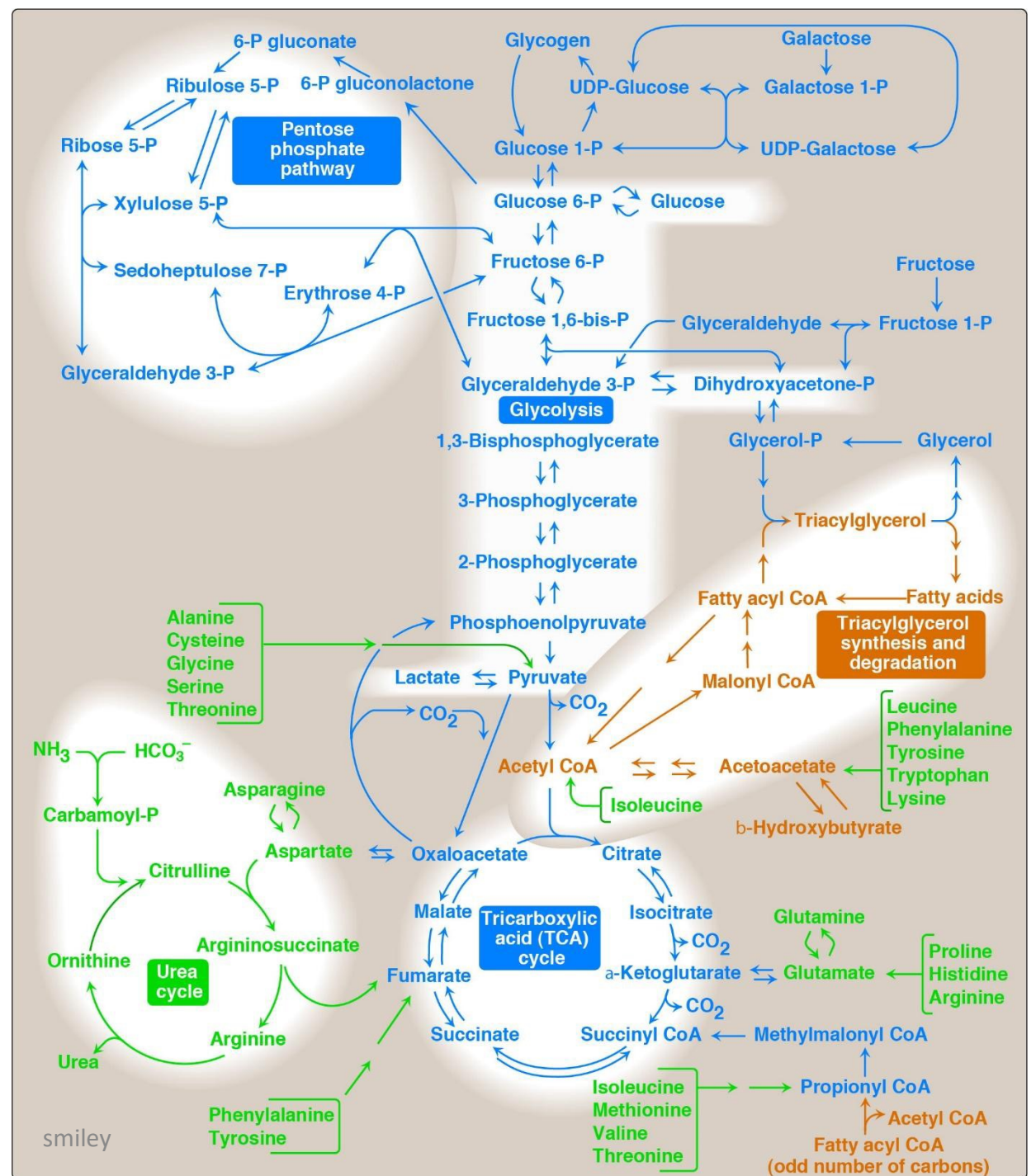
Title: Adipose tissue lipolysis; Fatty acid oxidation

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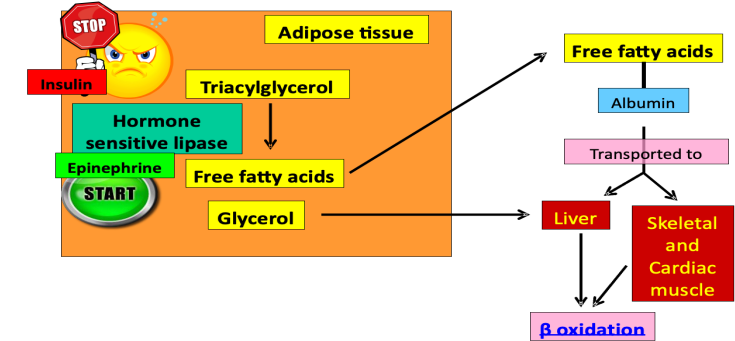
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Intermediary metabolism (Metabolic map)



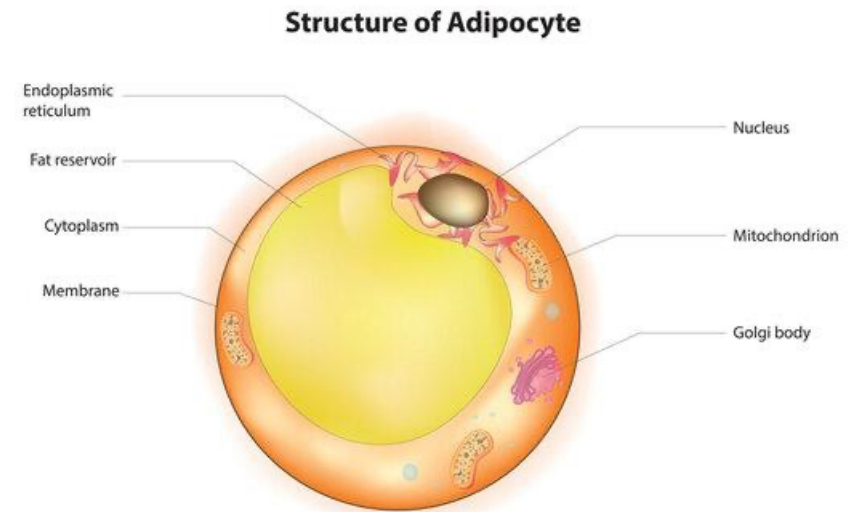
Lipid mobilization from Adipose tissue (Lipolysis)

- TAG = is storage form of Lipids
- TAGs are stored in Lipid droplets in **White Adipose tissue**
- FAs in TAG released as FFAs (non-esterified fatty acids) by HSL
- Glycerol formed by lipolysis, **NOT** reused in Adipose, **DOES NOT** have Glycerol Kinase
- Glycerol goes to Liver, to enter Glycolysis or Gluconeogenesis or TAG synthesis



HSL Regulation

- Inhibited by insulin (inactive dephosphorylation state)
- Low levels of insulin during fasting, stimulates HSL
- Epinephrine stimulates HSL (active in Phosphorylated state)

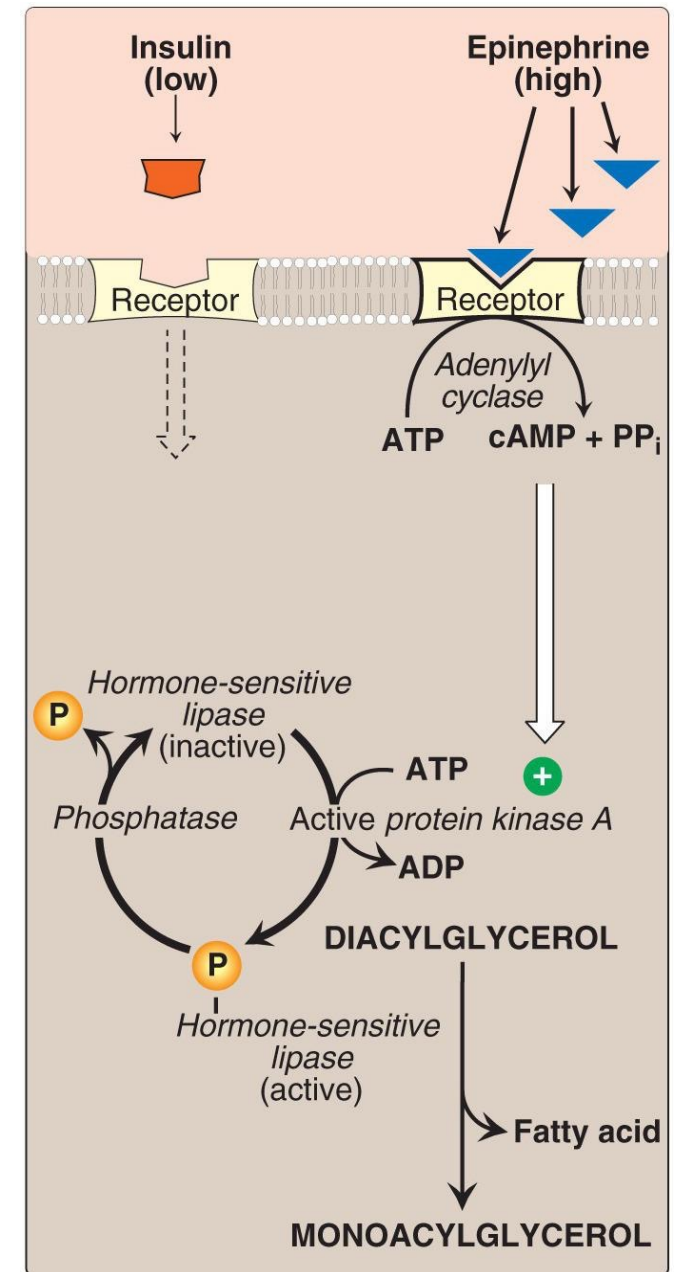


Epinephrine = Fasting, Fight or Flight

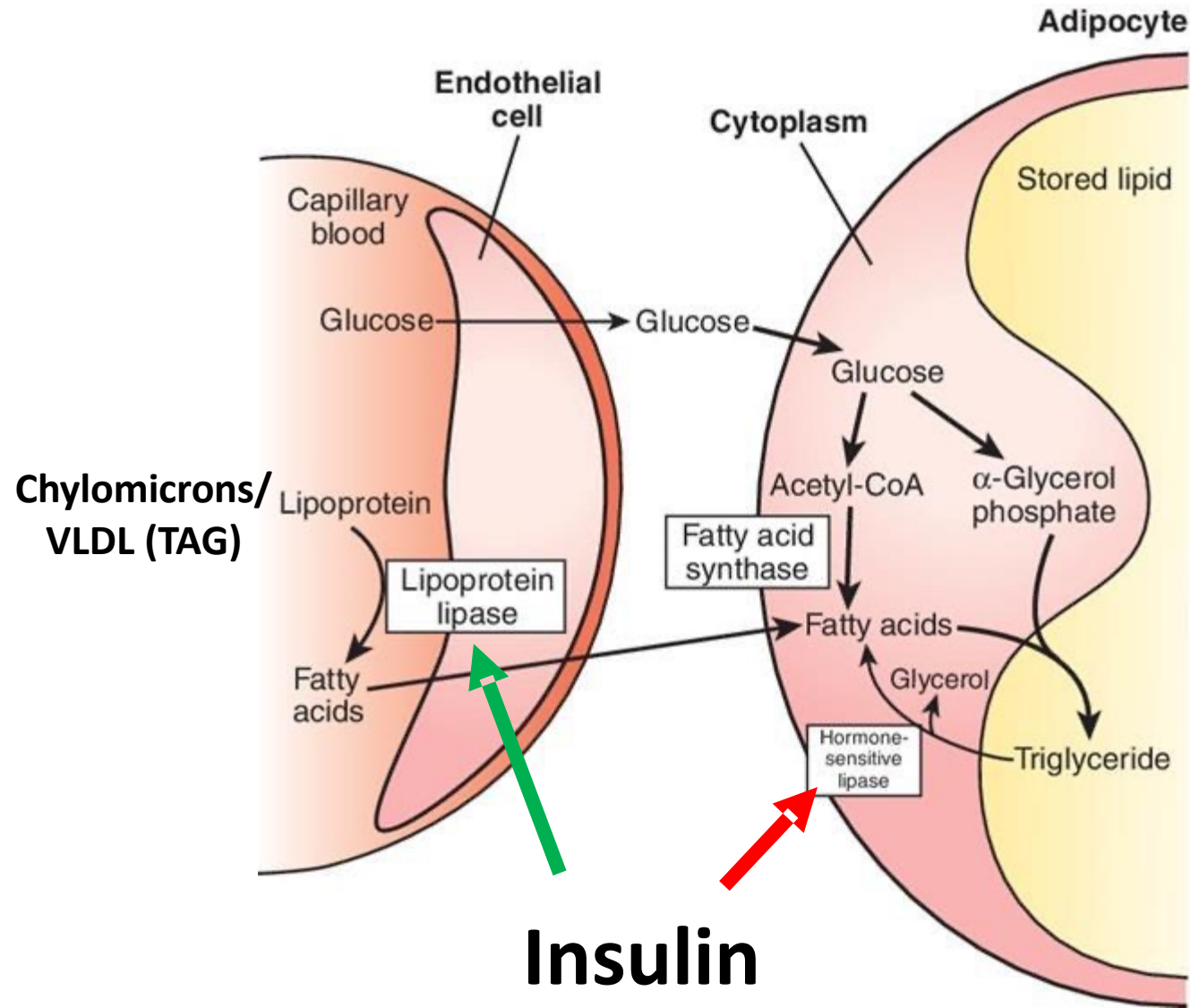
- Epinephrine **activates PKA**
- PKA **Phosphorylates** of HSL
- HSL is **active** in Phosphorylated state
- HSL **hydrolyzes TAG** into = 3 FAs + Glycerol

Insulin Resistance & Diabetes Mellitus

- Explain changes in type 1 and type 2 diabetes
- **Elevated serum FFAs = indicator of Insulin-resistance**



Compare HSL & LPL



HSL vs. LPL vs. Pancreatic Lipase

HSL

- Location: Within **Adipocytes**
- Role of insulin: **Insulin inhibits HSL**
- Action: Breaks down storage **TAG** in adipose tissue **to FFAs & Glycerol**
- **Lab marker** for increased HS lipase activity
Elevated serum FFAs
- **Increased HSL activity** seen in
 - Starvation
 - Insulin-resistance (Metabolic syndrome)
 - **Uncontrolled type 1** diabetes mellitus **HSL activity** markedly **increased**

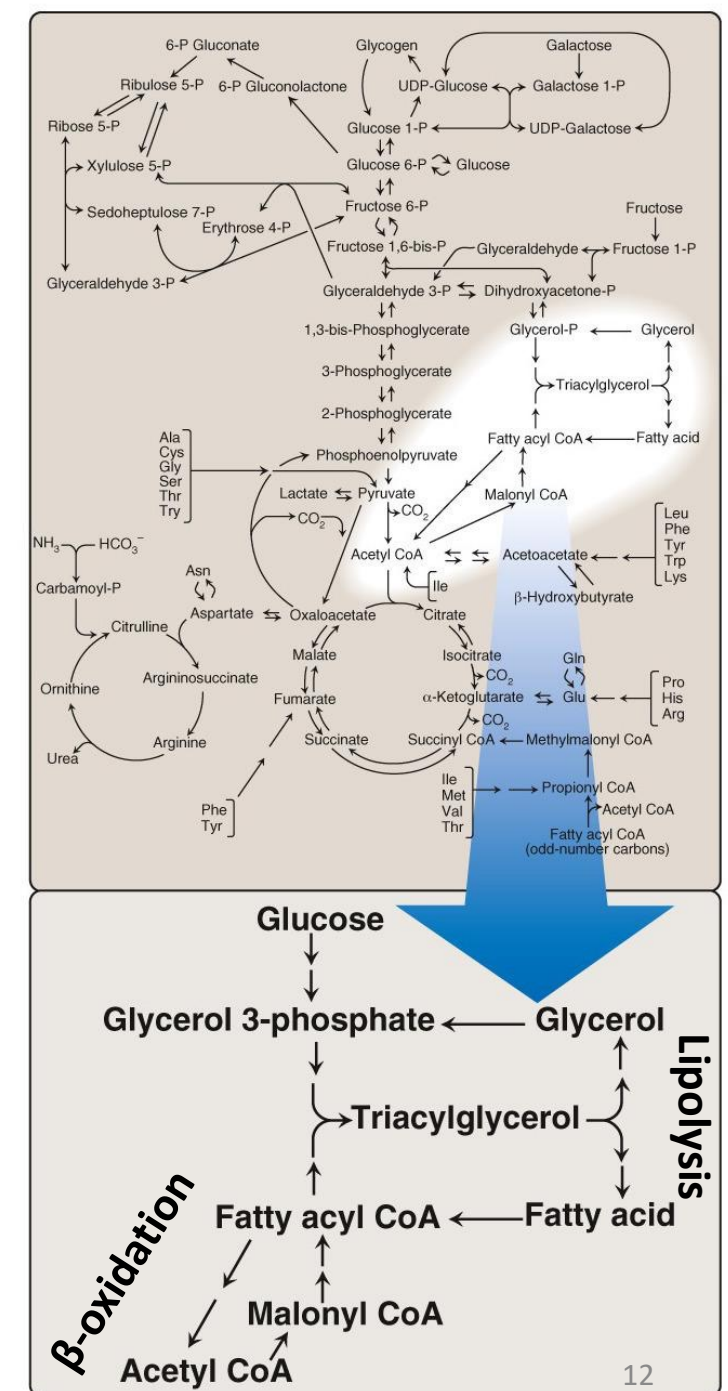
LPL

- Location: **Endothelium** of blood **vessels in Adipose** tissue & **Muscle**
- **Insulin induces LPL** for required optimum activity
- Action: **breaks down TAGs in Lipoproteins (Chylomicrons & VLDL) to FFA** that **enter Adipocytes** = to **form TAG** for **storage**
- **Lab** marker for low lipoprotein lipase activity:
Increased TAG & Increased VLDL
- **Decreased LPL activity** seen in
 - **Insulin resistance (Metabolic syndrome)**
 - **Type 2** diabetes mellitus – this results is **increased** serum **TAG (VLDL)** – which **in turn increases** the **risk** for **cardiovascular** disease (**macrovascular** complications)

Pancreatic Lipase = Enzyme for **digestion** of dietary TAG

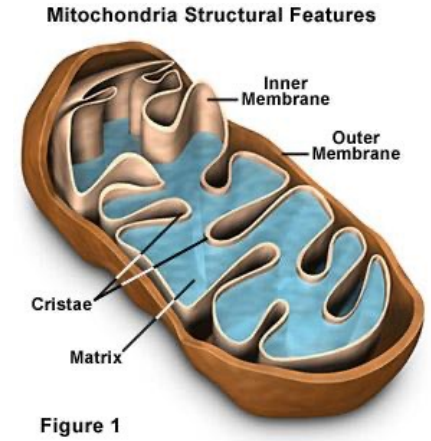
Overview of β -oxidation

- **FFAs are bound to albumin** in circulation
- **FFAs transported to Liver & Muscle (Skm and Cardiac)** to be utilized for **β -oxidation (AEROBIC)**
- The **brain DOES NOT** do **β -oxidation**, even during prolonged starvation, **Brain relies on Glucose & KBs**
- **β -oxidation CANNOT** take place in cells that lack **Mitochondria** or **Oxygen**
- **β -oxidation** occurs in the **Mitochondria**



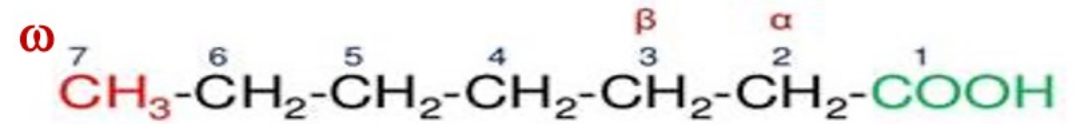
β -oxidation

- Oxidation of FAs @ β -C atom of fatty acid
- Intracellular location = Mitochondria



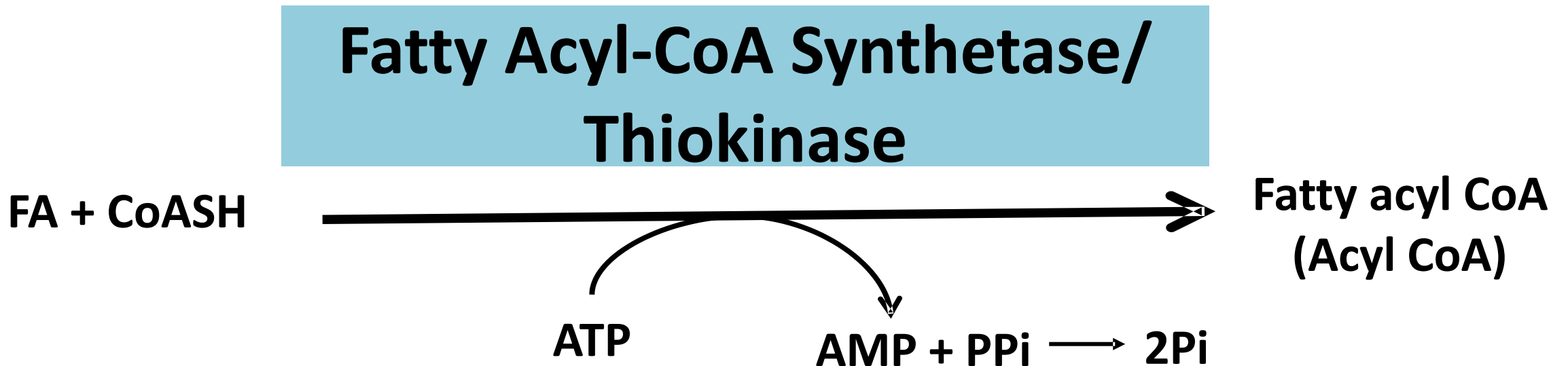
Stages of β -oxidation

1. Activation of FAs in the Cytosol
2. Transport ACTIVATED FA from cytosol to Mitochondria via the Carnitine shuttle
3. β -oxidation proper RXNS



Step 1 = Activation of FA in Cytosol

- FA is activated to Fatty Acyl-CoA by Fatty Acyl-CoA Synthetase (Thiokinase)
- Enzyme is on the Outer Mitochondrial Membrane (Cytosolic side)



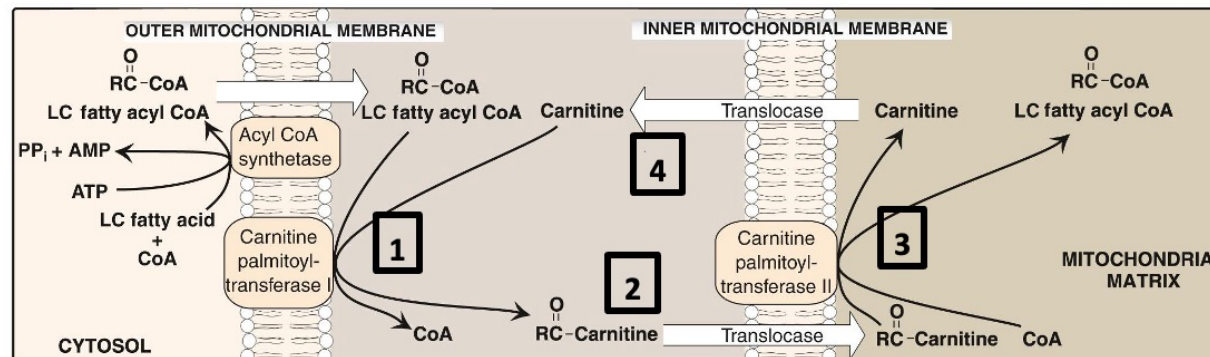
Step 2 = Carnitine Shuttle i.e. Transport of FA from Cytosol to Mitochondria

- **CPT-I** and **CPT-II** in **Outer** and **Inner** leaflet of **Mitochondrial membrane**, respectively
- **CPT-I** and **CPT-II** exist as different **isoforms** in the **Liver** and **Muscle**

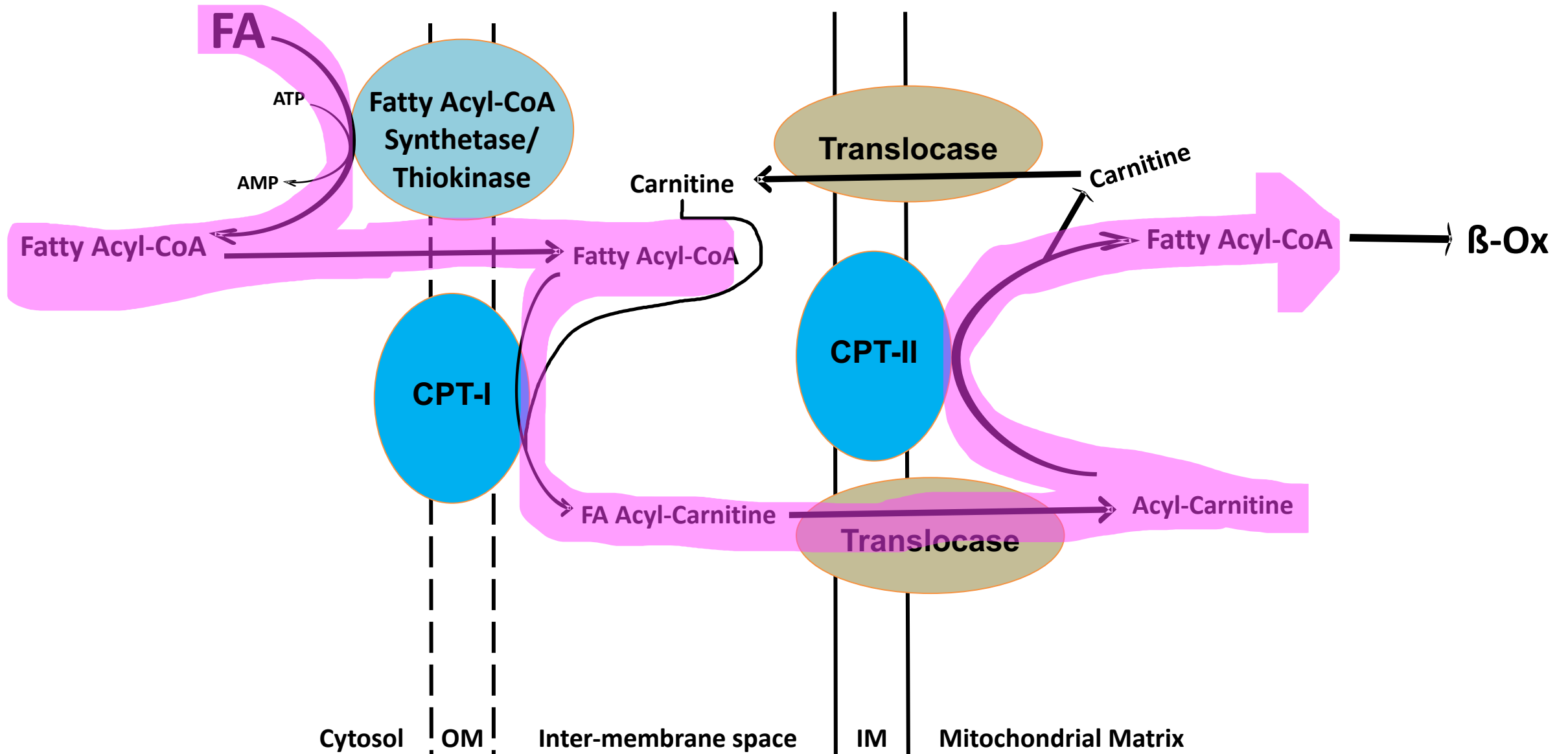
Fatty Acyl-CoA CANNOT enter the inner mitochondrial membrane, therefore must be shuttled

1. **CPT-1** adds **Carnitine** to **Acyl-group** in order to form = **FA Acyl-Carnitine**
2. **FA Acyl-Carnitine** transported **across** inner mitochondrial membrane **via translocase**
3. **Fatty Acyl-CoA** is then **re-formed** in the **matrix** by **CPT-II** and **can now enter the β -oxidation** pathway in Mito
4. **Carnitine** is transported **back to intermembrane space via translocase** to be used again

Regulation: **Malonyl-CoA** (intermediate of FA Synthesis) **inhibits CPT-I** to prevent futile cycle



Carnitine Shuttle



Reactions of β -oxidation

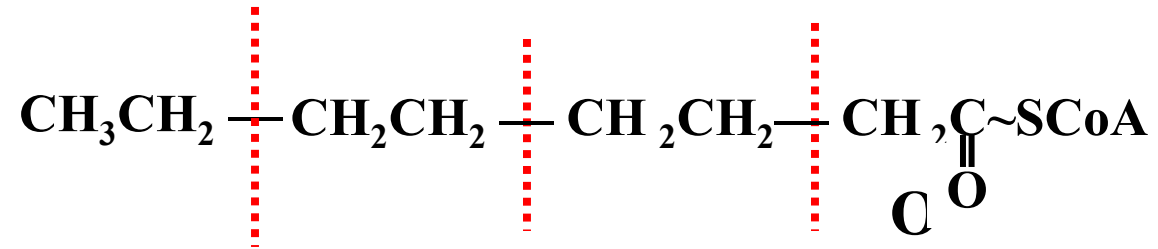
- Enzymes in mitochondrial matrix

RXN Sequence of β -oxidation

1. Oxidation (removal of H) (requires FAD)
 2. Addition of water
 3. Oxidation (removal of H) (requires NAD⁺)
 4. Cleavage
- 1 cycle of 4 rxns = results in cleavage of 2 C-atoms removed as Acetyl-CoA

Overview of Beta Oxidation

8 carbon Fatty Acid Acyl-CoA

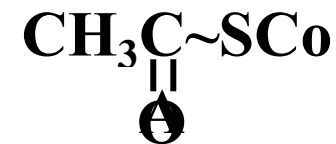
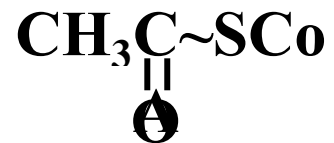
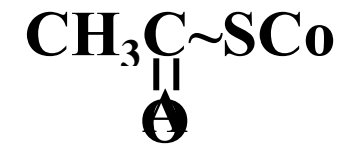
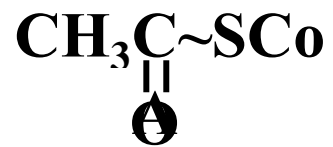


4 two carbon Acetyl-CoAs

Overview of Beta Oxidation

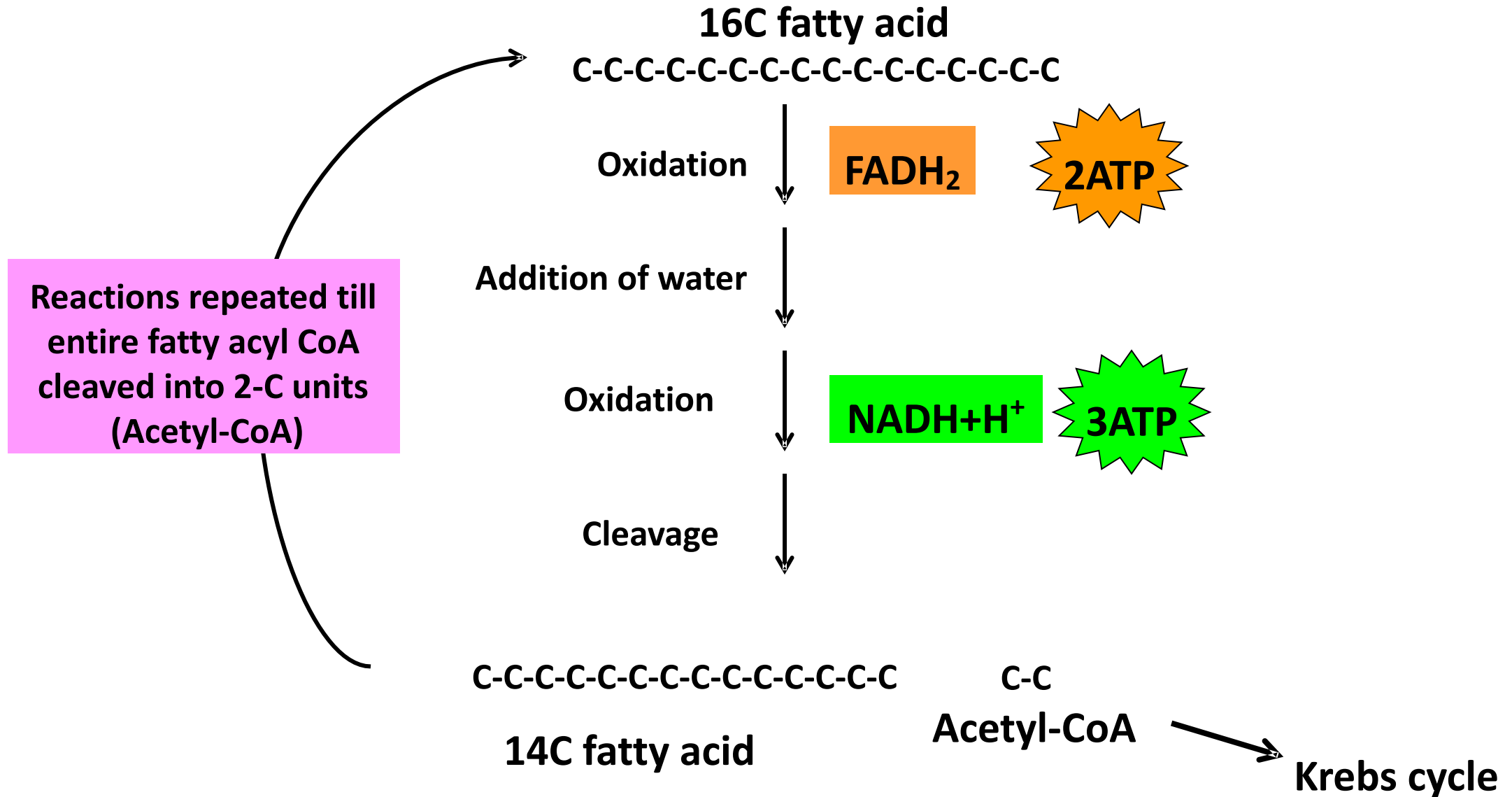
8 carbon Fatty Acid (Acyl-CoA)

3 rounds / 3 cuts



4 two carbon Acetyl-CoAs

Reactions of β -oxidation



Enzymes of β -Oxidation

Step 1 = **Acyl-CoA Dehydrogenase (FAD) - Oxidation**

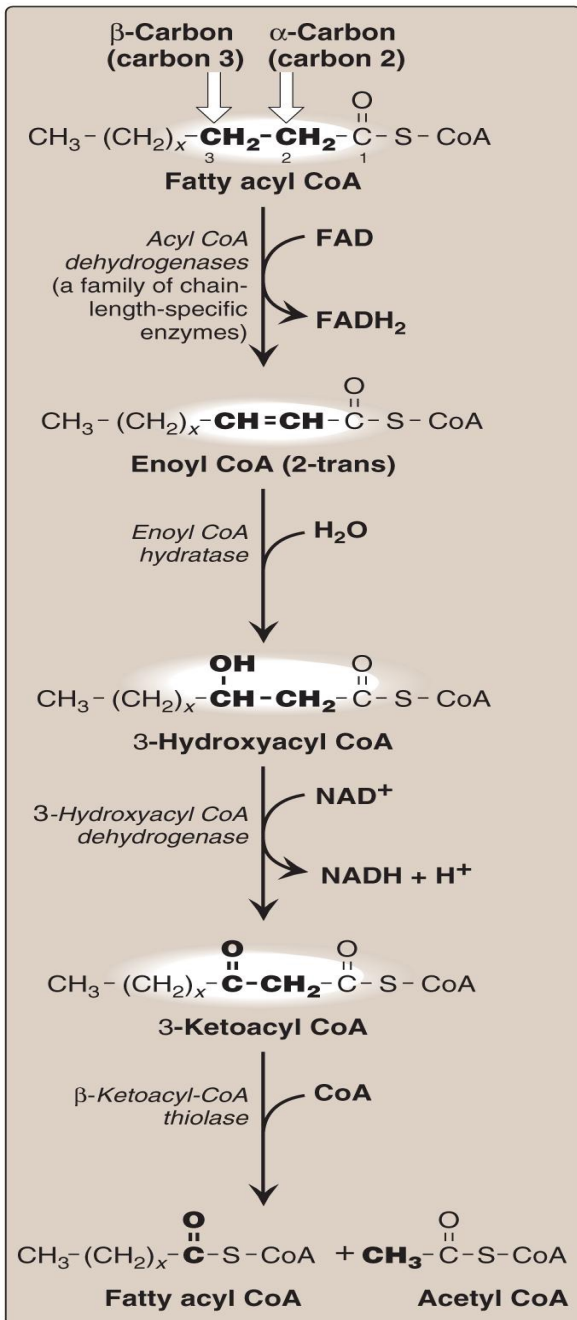
Step 2 = **Enoyl-CoA Hydratase** - Addition of **Water**

Step 3 = **3-Hydroxy Acyl-CoA Dehydrogenase (NAD⁺) - Oxidation**

Step 4 = **Thiolase** – performs the **Cleavage**

AE3T

β-Oxidation



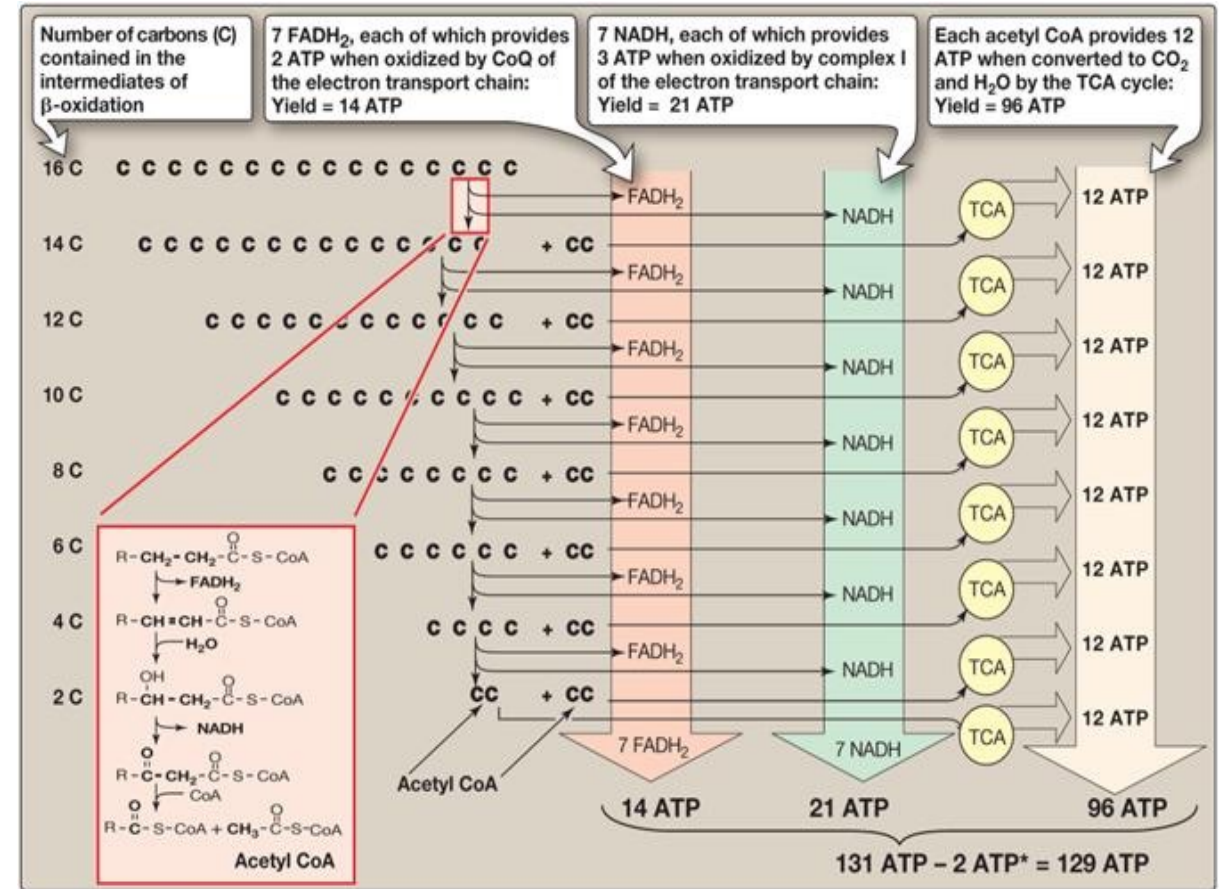
Reactions of β -Oxidation on Palmityl CoA

One cycle of β -Oxidation produces

- FADH₂
- NADH
- Acetyl-CoA

Complete β -Ox of 16C Palmitic acid

- 7 FADH₂
- 7 NADH
- 8 Acetyl CoA



- Acetyl-CoA enters TCA cycle to produce 12 ATP

TOTAL Energy yield from β -oxidation of Palmitic acid

Energy Value

- **1 Acetyl-CoA through TCA = 12 ATP**
- **1 FADH₂ = 2 ATP**
- **1 NADH = 3 ATP**

1 Round of β -Oxidation

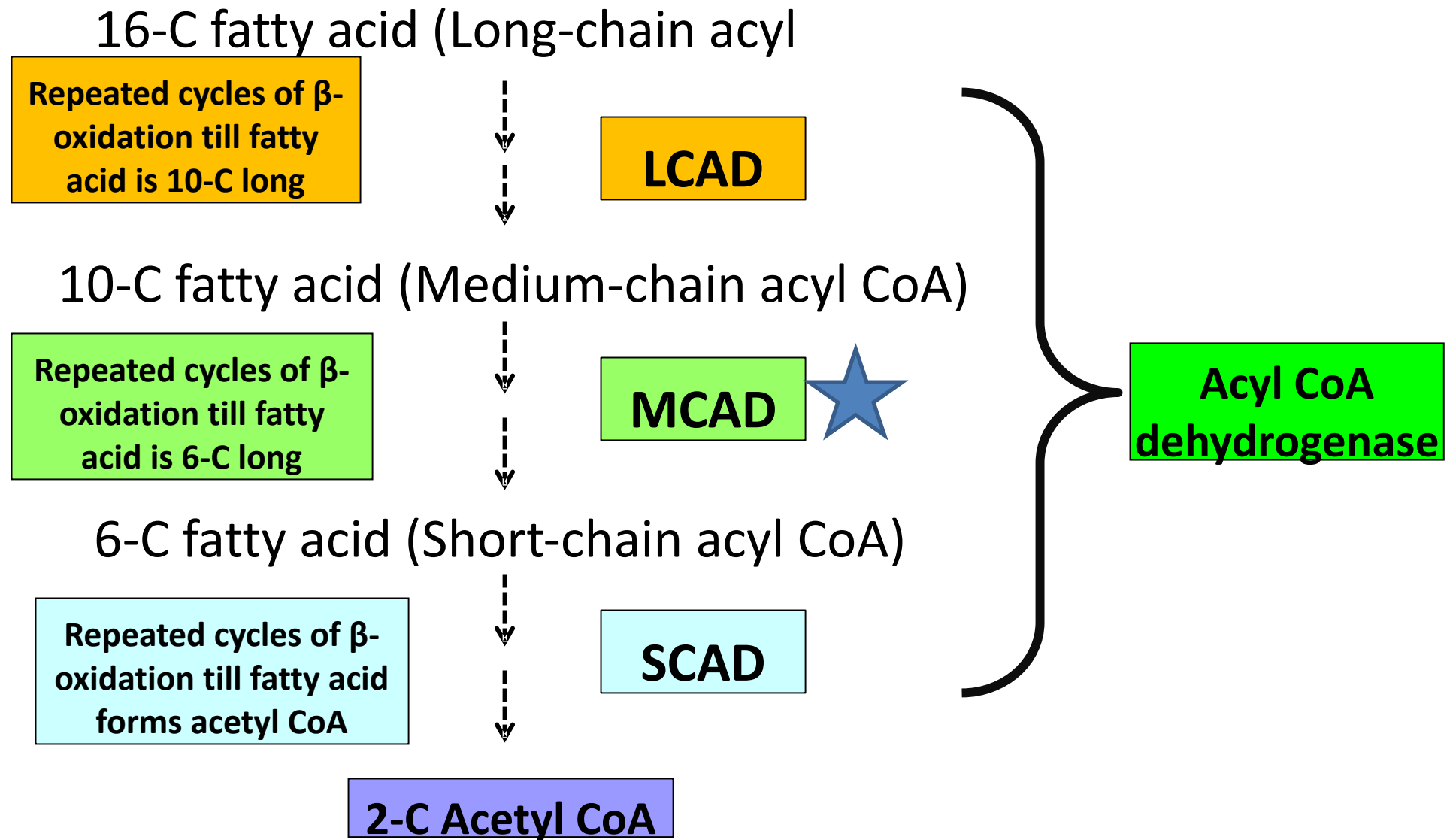
- **1 Acetyl-CoA = 12 ATP**
- **1 FADH₂ = 2 ATP**
- **1 NADH = 3 ATP**

*** -2 ATP for FA activation***

- **β -oxidation of 16C Palmitic acid = 8 Acetyl CoA, 7 FADH₂, 7 NADH**
- **8 Acetyl-CoA x 12 = 96 ATP**
- **7 FADH₂ x 2 = 14 ATP**
- **7 NADH x 3 = 21 ATP**
- **Total = 131 ATP, then -2 ATP for activation of FA (Step 1 of Carnitine Shuttle, used by Fatty Acyl-CoA Synthetase)**
- **Net ATP formed = 129 ATP**

- Compare to Oxidation of Glucose (38 ATP per glucose molecule)

Beta-Oxidation palmitic acid: **Acyl CoA dehydrogenase**: LCAD, **MCAD** and SCAD

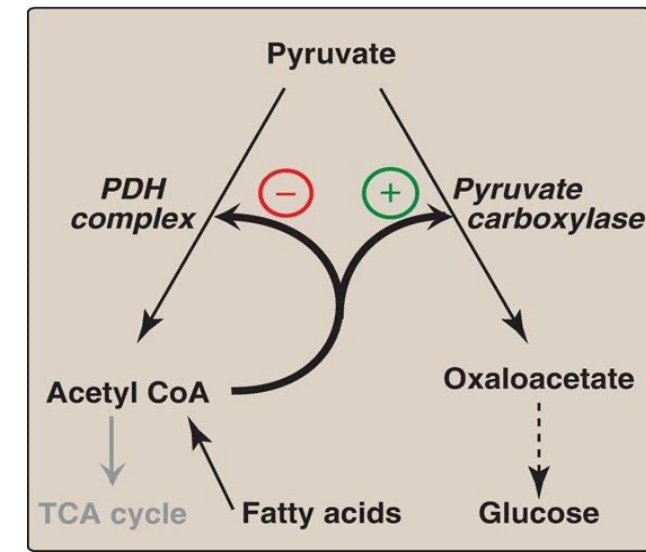


β -Oxidation in Liver

Overnight fast, 60-70% energy from β -oxidation

β -oxidation in Liver is important for

- Gluconeogenesis
- Ketone Body Synthesis



Systemic β -Oxidation Disorders = MCAD deficiency, Primary Carnitine deficiency, CPT-I deficiency affects Liver, and manifests as

– **Hypoglycemia** = cells rely on Glucose for energy & reduced Gluconeogenesis, Acetyl CoA absolute activator for Pyruvate Carboxylase (Gluconeogenesis)

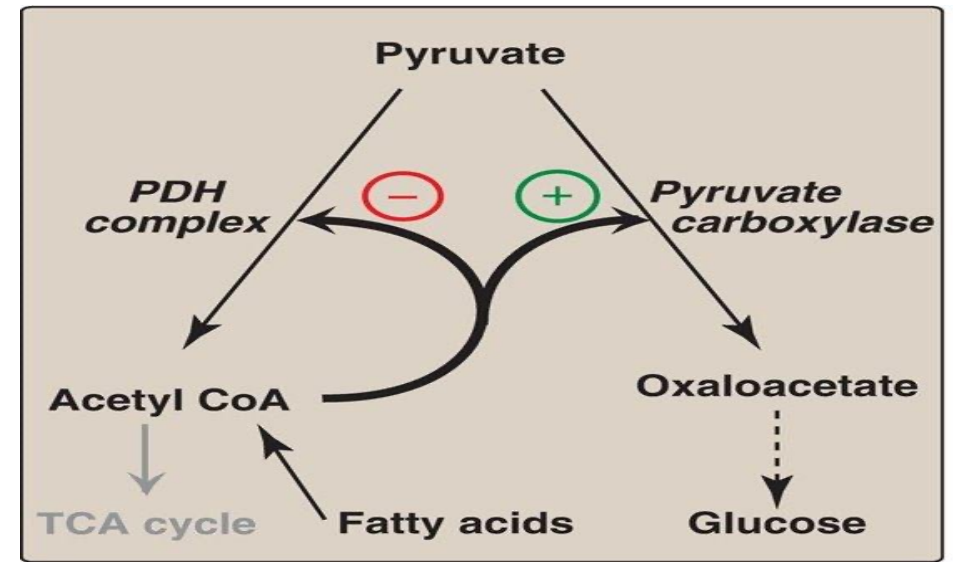
– **Hypoketosis** = reduced production of Acetyl-CoA

Case report: MCAD deficiency

- An 18-month-old girl brought to emergency room because she became unresponsive. <https://www.youtube.com/watch?v=p9APKkv8RC0>
- Episodes of **hypoglycemia** ever since she was 11 months old
- Episodes occurred on fasting (>8 hours) or **following illness**
- On admission (metabolic crisis),
 - **Plasma glucose is 38mg/dL (Hypoglycemia)**
 - Free fatty acids in blood are elevated (normal), but
 - **Ketone bodies are below normal (Hypoketosis)**
 - **C8-C10 acyl carnitines elevated**
- **Differentiate from von Gierke disease**

MCAD deficiency

- Systemic β -Oxidation Disorder
- **Autosomal Recessive** disorder (newborn screen)
- Age of presentation = 6 -24 months



Severe Hypoglycemia (Metabolic Crisis) while fasting or illness

- Liver & Muscle **CANNOT** use FAs for energy, Acetyl-CoA **CANNOT** activate PC
- **Glucose** is the **sole source** of energy, **resulting** in profound hypoglycemia
- **Impaired Gluconeogenesis** = **Less ATP** and **Less Acetyl-CoA**
- **Liver dysfunction** and **elevated AST** and **ALT** levels **during crisis**
- **Decreased oxidation of Medium Chain FAs (8-10 C atoms)** = **Less Acetyl CoA** = **Less KB synthesis**

MCAD deficiency - Systemic β -Oxidation disorder

Lab findings

- **Hypoglycemia**
- **Hypoketosis**
- **Medium-chain Acyl-Carnitines in urine**
- **Dicarboxylic acids in urine (increased ω -oxidation)**

Management

- **Manage Hypoglycemia using IV Glucose**
- **Frequent feeding = high carbohydrate, low-fat diet**
- **Avoid fasting, monitor during illness**
- **Carnitine supplements**

Carnitine deficiency – Systemic β -Oxidation disorder

- **Defective transport of Long-chain FAs into Mitochondria ~ β -oxidation impaired**

Primary or Systemic Carnitine deficiency

- Presents at an **early age** of presentation
- **IMPAIRED Carnitine UPTAKE into tissues**
- **Hypoglycemia due to impaired Gluconeogenesis (Acetyl-CoA low)**
- **Hypoketosis**
- **Liver dysfunction and Elevated AST and ALT**
- **Cardiomyopathy**

Secondary Carnitine deficiency

- **Disorders like MCAD = Loss of Acyl-Carnitine in urine**

Management

- **Avoid fasting; monitor during illness**
- **Carnitine supplements**

CPT-I deficiency – Systemic β -Oxidation disorder

- **Hypoglycemia and Hypoketosis**
- **Affects Liver isoform (systemic form)**

Management

- **Avoid fasting; monitor during illness**
- **Carnitine supplements**

Compare Liver GSDs & Systemic β -oxidation Disorders

Systemic β -oxidation Disorders

- HYPOGLYCEMIA
- Hypoketosis
- Age: 6 months-2years
- Liver biopsy = Excessive Lipid droplets (TAG)
- Elevated liver enzymes (ALT and AST)
- MCAD/CPT-I/Primary Carnitine -deficiency

Liver GSD (von Gierke disease)

- HYPOGLYCEMIA
- KB levels increased
- Age: 6 months - 2years
- Lactic acidosis
- Hypertriglyceridemia; hyperuricemia
- Hepatomegaly
- Liver biopsy: Excessive Glycogen accumulation (structure normal)
- Elevated liver enzymes (ALT and AST)
- 'Doll' face
- Glucose 6-phosphatase deficiency

β-oxidation in Skeletal Muscle

- **Skm uses β-oxidation during episodes of Endurance/ Aerobic exercise** (low-intensity prolonged activity)
- **Provides energy for sustaining the AEROBIC exercise**

Myopathic β-oxidation disorders

- **Defect in β-oxidation in Skm & Cardiac muscle**
- **Manifests as Cramps during low-intensity, sustained, aerobic activity** (GSDs cramp during Anerobic)
- **Unable to perform aerobic exercise**

Lab findings

- **Forearm Exercise Test = blood Lactate levels increase following ischemic exercise** (Indicates normal anaerobic metabolism)
- **Myoglobinuria, and elevated levels of serum CK-MM**
- **Muscle biopsy shows = shows Lipid droplet accumulation in Muscles**

CPT-II deficiency - Myopathic β -oxidation disorders

Cardiomyopathy & Muscle weakness (Myopathic form)

- **Unable to tolerate aerobic activity** (Cramps during aerobic exercise)
- **Lipid TAGs** deposited in **Sk**m
- **Forearm Exercise Test = Lactate increase**
- **Myoglobinuria** and elevated serum **CK-MM** levels
- Affects muscle isoform

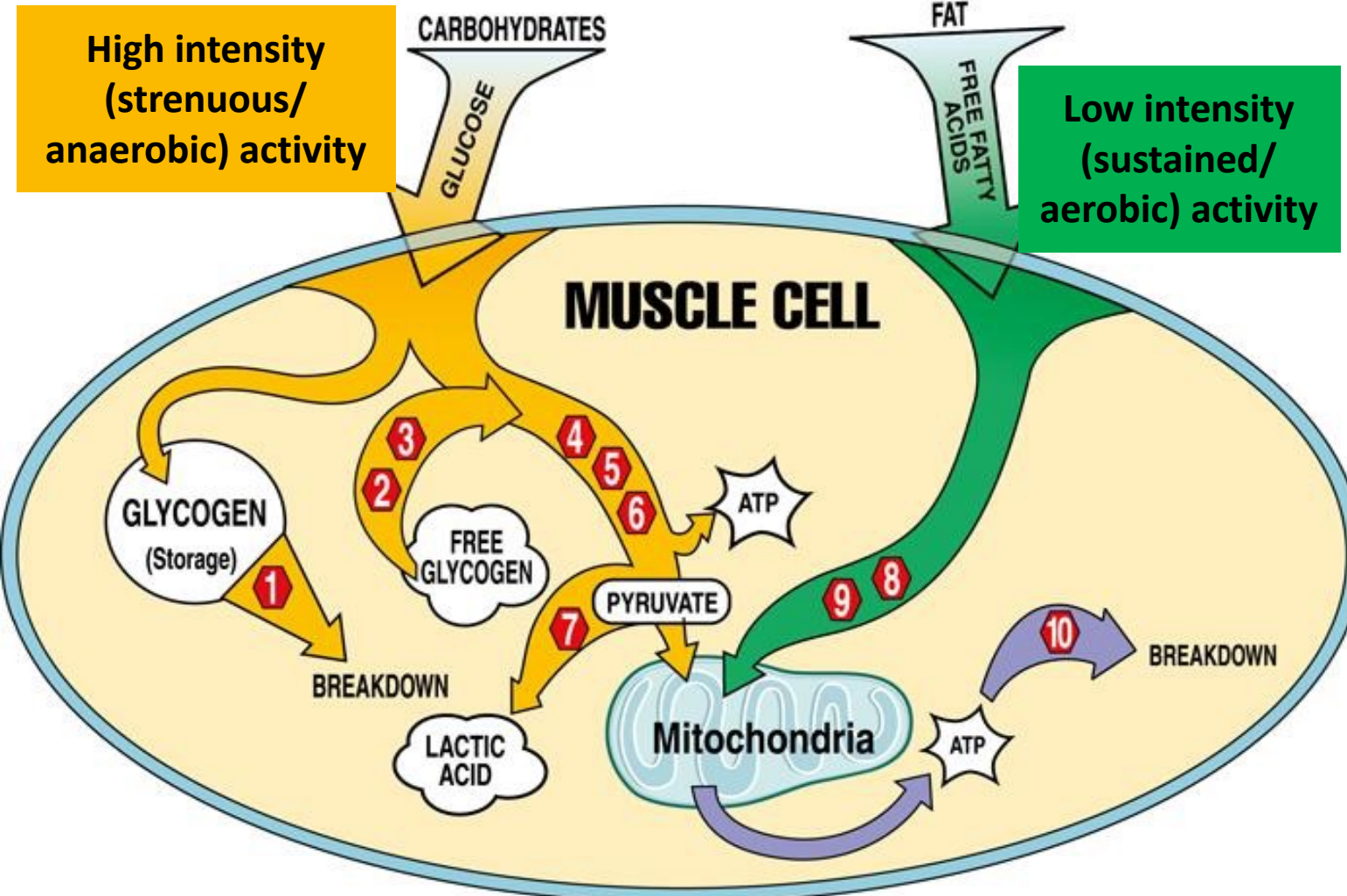
Management

- **Cease muscle activity** and **give Glucose**
- **Carnitine supplements**

Metabolic myopathies

<https://www.mda.org/disease/metabolic-myopathies/types>

Roadblocks Lead to Disorders



Muscle glycogen storage disorder
Muscle phosphorylase deficiency (McArdle)
Phosphofructokinase deficiency (Tarui)

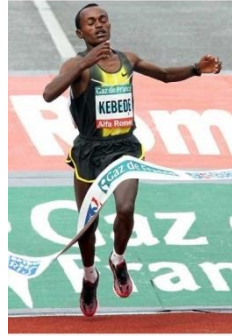
Myopathic fatty acid oxidation defect
Carnitine palmityl transferase-II deficiency

Skeletal muscle depends on energy from carbohydrates and fats. Fuels stored in muscle (glycogen) or imported directly from bloodstream (glucose and fatty acids)

Differentiate between muscle glycogen storage disorder and myopathic fatty acid oxidation defect

Aerobic and anerobic exercise: Fuel sources

Aerobic exercise



- Oxygen present; Abundant mitochondria
- Glycogenolysis is active
- Glycolysis is active
- Kreb's cycle and electron transport chain are both active, producing ample ATP
- **Fatty acid oxidation**
- ATP production is maximal
- Muscle fatty acid oxidation defect → Myopathic beta-oxidation defect → Inability to tolerate aerobic activity

From DM lecture 1

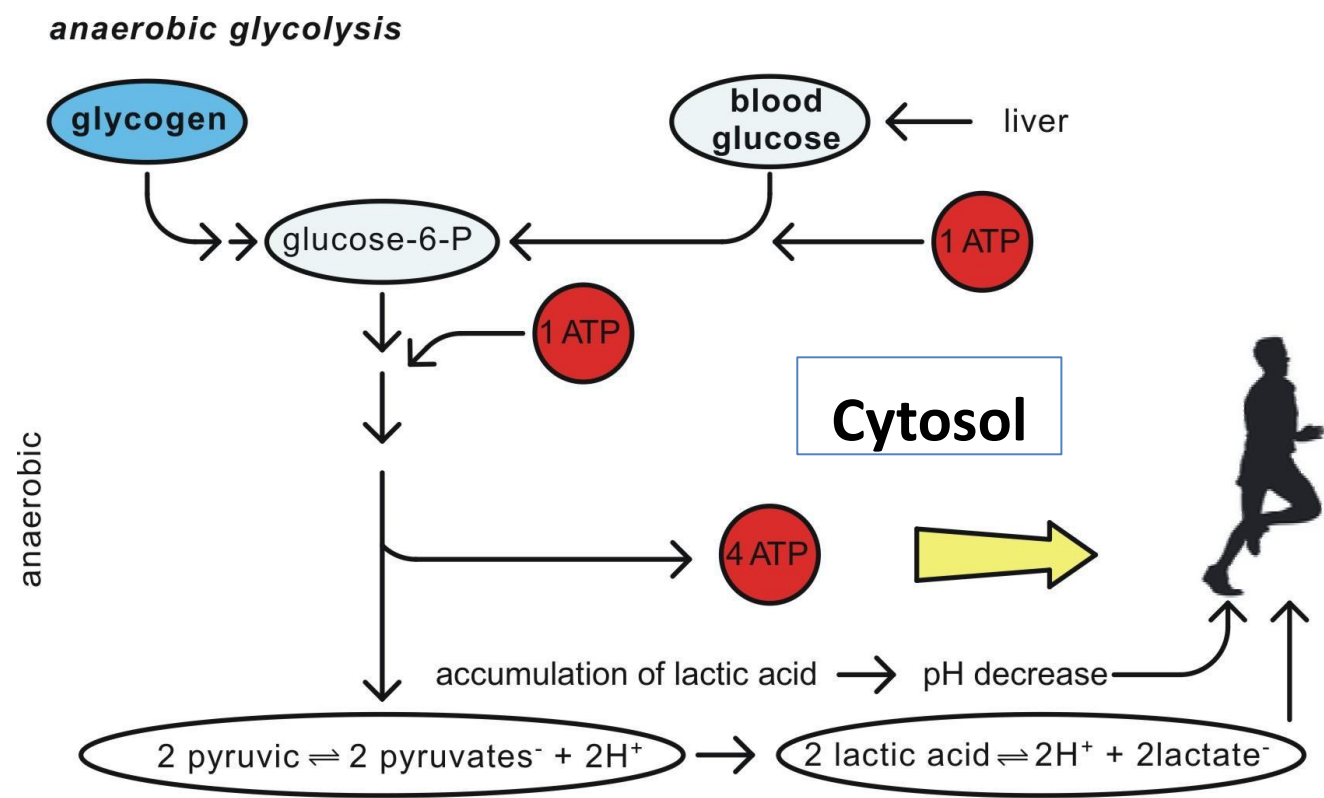
Anaerobic/ strenuous/ ischemic exercise



- Oxygen absent; Cytosol
- **Glycogenolysis is active, but of short duration**
- **Glycolysis is active**
- **Pyruvate reduced to LACTATE**
- ATP production is rapid but minimal
- Defect in Muscle Glycogenolysis/ glycolysis → Muscle glycogen storage disorder → Inability to tolerate anaerobic activity

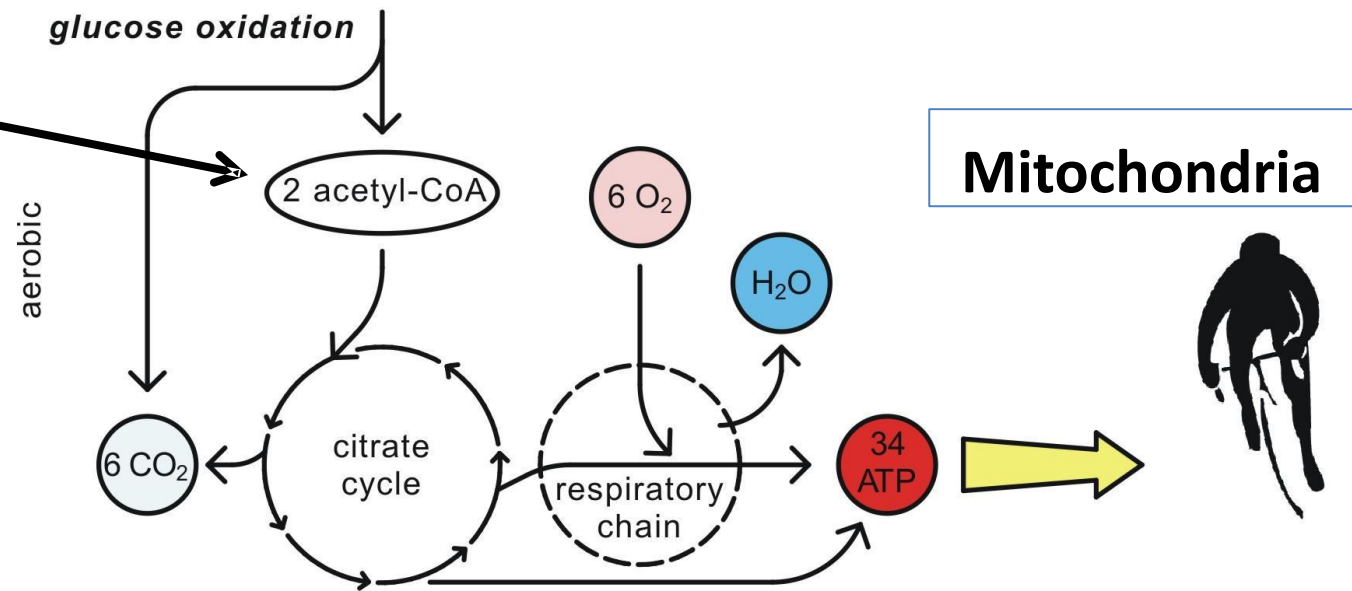
Metabolism in muscle: Aerobic and anaerobic

Glycogenolysis and Glycolysis (Anaerobic)



Aerobic: Fatty acid oxidation and Krebs cycle

Fatty acid oxidation



β-oxidation Disorders

Systemic β-oxidation Disorders

- Age: **6 months-2 years**
- Organs affected: **Liver and Muscle**
- **Hypoglycemia and Hypoketosis (metabolic crisis) following an illness**
- Seen in: **MCAD / CPT-I / Primary Carnitine -deficiency**

Management

- **Prevent hypoglycemia**
- **Carnitine supplements**

Differentiate from vonGierke disease

Myopathic β-oxidation disorders

- Age: **Adolescents/ Young adults**
- Organs affected: **Muscle (Skeletal and Cardiac)**
- Muscle **cramps** during **aerobic exercise**
- **Lab tests: Myoglobinuria** and increased **CK-MM** levels
- Muscle **biopsy: Elevated muscle TAGs in Muscle**
- **Forearm Exercise Test = Lactate increase**
- Seen in **CPT-II deficiency**

Management

- **Carnitine supplements**

Differentiate from McArdle disease



Jamaican Vomiting Sickness



- Eating **Unripe Ackee fruit** (Caribbean and Africa)
- Results in **Hypoglycemia & Vomiting** ~ symptoms occur **2-6 hours after**
- **Hypoglycin A** in the fruit= **inhibits MCAD (β -oxidation)**
- **Inhibition of β -oxidation leads to profound Hypoglycemia** due to **less Acetyl-CoA available to activate Gluconeogenesis (activates PC)**
- **Drowsiness (Hypoglycemia), coma and death**
- **Medium-chain Acyl-Carnitines found in urine**

Unripe ackee fruit



Regulation of β -oxidation

Serum FFAs levels regulate rate of β -oxidation

- **Low** insulin/glucagon ratio (fasting; Type-1 Diabetes Mellitus), **activates HSL**
- **High** insulin/glucagon ratio (fed state) **inhibits HSL**

Entry of FAs into Mitochondria by CPT-I

- **Malonyl-CoA** from FA Synthesis **inhibits CPT-I**
- β -oxidation & FA Synthesis do not occur simultaneously (Reciprocal regulation)

Compare Muscle GSD & Myopathic β -oxidation Disorders

Myopathic β -oxidation Disorders

- **Cramps** during **Aerobic** exercise
- Age: **Adolescents/ Young** adults
- **Blood lactate** levels **increase** following ischemic/ **anaerobic** exercise (**normal** test)
- **Muscle biopsy: Excessive Lipid droplets (TAGS)**
- **Myoglobinuria** and elevated serum **CK-MM**
- **CPT-II deficiency**

Muscle GSD (McArdle disease)

- **Cramps** during **ANAerobic** exercise (**strenuous** exercise)
- Age: **Adolescents/ Young** adults
- **NO** increase in **blood lactate** following ischemic /**anaerobic** exercise (**abnormal**)
- **Muscle biopsy: Excessive Glycogen**
- **Myoglobinuria** and elevated serum **CK-MM**
- **McArdle**: Muscle glycogen phosphorylase deficiency

Comparison of synthesis and degradation of long chain even numbered fatty acids

	SYNTHESIS	DEGRADATION
Greatest flux through pathway	After carbohydrate-rich meal	In starvation
Hormonal state favoring pathway	High insulin/glucagon ratio	Low insulin/glucagon ratio
Major tissue site	Primarily liver	Muscle, liver
Subcellular location	Primarily cytosol	Primarily mitochondria
Carriers of acyl/acetyl groups between mitochondria and cytosol	Citrate (mitochondria to cytosol)	Carnitine (cytosol to mitochondria)
Phosphopantetheine-containing active carriers	Acyl carrier protein domain, coenzyme A	Coenzyme A
Oxidation/reduction coenzymers	NADPH (reduction)	NAD ⁺ , FAD (oxidation)
Two-carbon donor/product	Malonyl CoA: donor of one acetyl group	Acetyl CoA: product of β -oxidation
Activator	Citrate	
Inhibitor	Long-chain fatty acyl CoA (inhibits <i>acetyl CoA carboxylase</i>)	Malonyl CoA (inhibits <i>carnitine palmitoyltransferase-I</i>)
Product of pathway	Palmitate	Acetyl CoA
Repetitive four-step process	Condensation, reduction, dehydration, reduction	Dehydrogenation, hydration, dehydrogenation, thiolysis